Chapter 12

Metal-Oxide-Semiconductor FET (MOSFET)

12.1 Introduction

The **Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET)** is the cornerstone of modern electronics. While the BJT and JFET have their specific uses, the MOSFET is the dominant device in the semiconductor industry. It is the fundamental building block of microprocessors, memory chips, and power switching circuits.

Why is the MOSFET so important?

1. **Input Impedance:** It has an incredibly high input impedance ($10^{14}\Omega$), far exceeding even the JFET. This means it draws virtually **zero current** from the driving source.

2. **Scalability:** It is easier to manufacture and scale down to nanometer dimensions than BJTs, allowing billions of transistors to fit on a single chip.

3. **Power Efficiency:** In digital circuits (CMOS), it consumes power only when switching states.

Learning Objectives

By the end of this chapter, you should be able to:

- Describe the construction of Depletion-type (D-MOSFET) and Enhancement-type (E-MOSFET) devices.
- Explain the function of the Silicon Dioxide (SiO_2) insulating layer.
- Analyze the characteristic curves and apply the relevant current equations.
- Calculate bias points for E-MOSFET circuits (Feedback and Voltage Divider).
- Discuss the operation of CMOS inverters.

12.2 Construction and The Insulated Gate

The defining feature of a MOSFET is the **Insulated Gate**.

In a JFET, the Gate forms a PN junction with the channel, which must be kept reverse-biased to prevent current flow. In a MOSFET, the Gate is physically separated from the semiconductor channel by a very thin layer of **Silicon Dioxide (SiO_2)**.

- **Metal:** The Gate contact (historically Aluminum, now often Polysilicon).
- **Oxide:** The SiO_2 layer. It is a dielectric (glass), making it a perfect electrical insulator.
- **Semiconductor:** The Silicon substrate containing the Source, Drain, and Channel.

Since the Gate is insulated, no current can flow between the Gate and the Channel, regardless of the polarity of the Gate voltage. This gives the MOSFET its ”Infinite” input impedance.

12.3 Depletion-Type MOSFET (D-MOSFET)

The Depletion-type MOSFET has a physical channel connecting the Source and Drain, even when no Gate voltage is applied.

12.3.1 Construction (N-Channel)

- Built on a P-type substrate.
- Two highly doped N-regions form the Source and Drain.
- A physical **N-Channel** connects the Source and Drain.
- The Gate is insulated from the channel by the oxide layer.

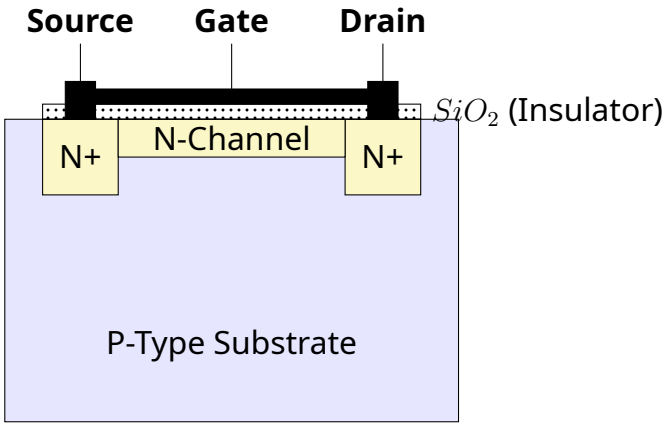


Figure 12.1: Cross-section of an N-Channel Depletion MOSFET.

12.3.2 Basic Operation

Because a physical channel exists, current (I_D) can flow even if $V_{GS} = 0$. The D-MOSFET is unique because it can operate in two modes:

Depletion Mode ($V_{GS} < 0$):

- When a negative voltage is applied to the Gate, the negative potential repels the free electrons (n-carriers) in the channel away from the oxide interface.
- This ”depletes” the channel of charge carriers.
- Resistance increases, Current (I_D) decreases.
- At a specific negative voltage (V_P or $V_{GS(off)}$), the channel is fully depleted, and current stops.

Enhancement Mode ($V_{GS} > 0$):

- Unlike JFETs (which cannot have forward bias), D-MOSFETs allow positive Gate voltage.
- A positive Gate voltage attracts *more* electrons from the P-substrate into the N-channel.
- Conductivity increases, and I_D rises *above* I_{DSS}.

12.3.3 Characteristics Equation

The D-MOSFET follows the same Shockley equation as the JFET:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2 \tag{12.1}$$

However, unlike the JFET, V_{GS} can be positive or negative.

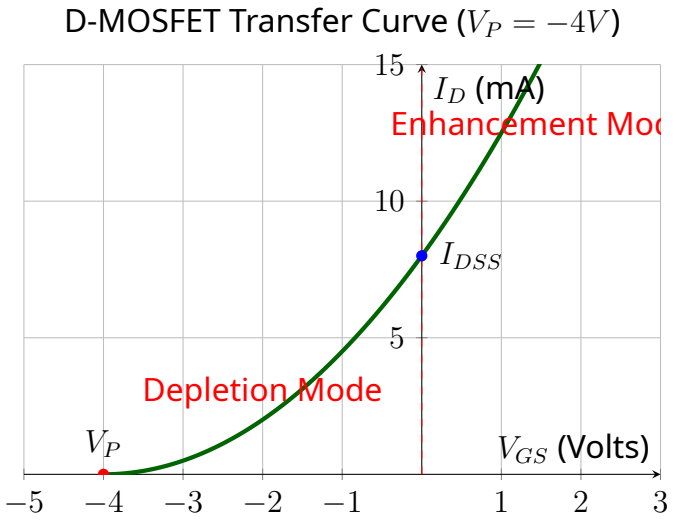


Figure 12.2: Transfer characteristics showing operation in both positive and negative regions.

12.4 Enhancement-Type MOSFET (E-MOSFET)

The E-MOSFET is the ”Normally OFF” device. It is the most widely used transistor in the world (digital logic).

12.4.1 Construction

- Similar to D-MOSFET, but there is **NO physical channel** connecting Source and Drain.
- The P-type substrate extends completely to the SiO_2 layer.
- With $V_{GS} = 0$, there is no path for electrons to flow. The resistance between Source and Drain is infinite.

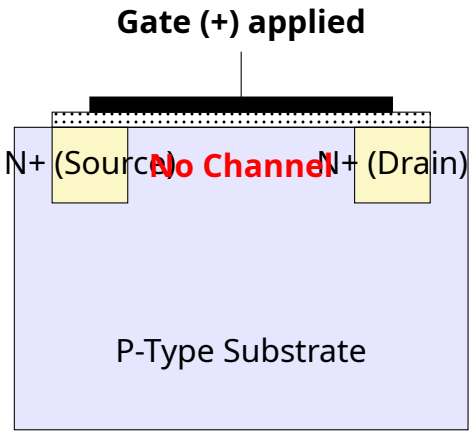


Figure 12.3: Structure of an E-MOSFET. Note the lack of a pre-existing channel.

12.4.2 Operation: Channel Induction

To turn the E-MOSFET ON, we must "enhance" the conductivity by creating a channel.

1. Apply a **Positive Voltage** to the Gate ($V_{GS} > 0$).
2. The metal Gate acts as the positive plate of a capacitor.
3. The electric field penetrates the SiO_2 and repels the holes (majority carriers) in the P-substrate away from the surface.
4. Simultaneously, it attracts **electrons** (minority carriers) from deep within the substrate towards the surface.
5. When enough electrons accumulate, they form a thin N-type layer connecting the Source and Drain. This is called the **Inversion Layer**.

12.4.3 Threshold Voltage (V_T)

The channel does not form instantly. A minimum amount of voltage is required to gather enough electrons to bridge the gap.

- **Threshold Voltage (V_T or $V_{GS(th)}$):** The minimum Gate-Source voltage required to create the inversion layer (turn the device ON).
- If $V_{GS} < V_T$, $I_D = 0$.
- If $V_{GS} > V_T$, I_D flows.

12.5 E-MOSFET Characteristics

12.5.1 Transfer Curve

The E-MOSFET does **not** follow Shockley's equation. Instead, the current is governed by:

$$I_D = k(V_{GS} - V_T)^2 \tag{12.2}$$

Where:

- k: A construction constant (A/V^2). Sometimes derived as $k = \frac{I_{D(on)}}{(V_{GS(on)} - V_T)^2}$ based on datasheet values.
- V_T: Threshold Voltage.

12.6 MOSFET Circuit Symbols

The symbols explicitly show the difference in internal structure.

- **D-MOSFET:** The channel line is **Solid**, representing the continuous physical channel.
- **E-MOSFET:** The channel line is **Broken** (Dashed), representing the normally open path.

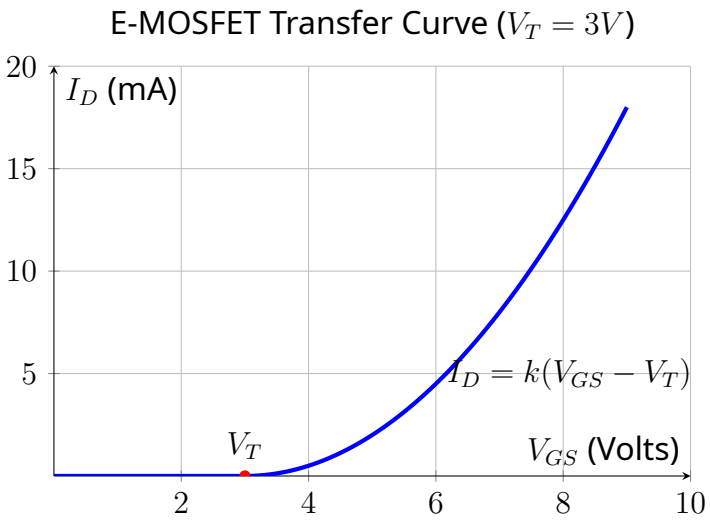


Figure 12.4: Transfer characteristics of an E-MOSFET showing no current until V_T.

- **Gap:** Notice the gap between the Gate and the Channel line, representing the oxide insulation.

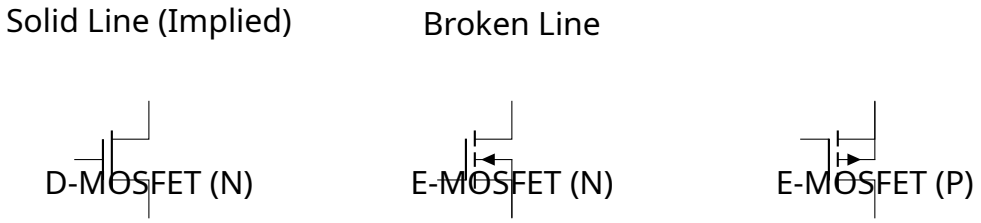


Figure 12.5: Standard Circuit Symbols for MOSFETs.

12.7 Biasing E-MOSFETs

Because E-MOSFETs require a positive Gate voltage ($V_{GS} > V_T$) to turn on (for N-channel), we cannot use the Self-Bias method used for JFETs (which produces negative V_{GS}).

12.7.1 Feedback Biasing

This simple method connects a resistor R_G from the Drain directly to the Gate.

- Since $I_G = 0$, there is no voltage drop across R_G.
- Therefore, $V_G = V_D$.

- Since Source is grounded, $V_{GS} = V_{DS}$.
- This ensures the MOSFET is always in the saturation (active) region.

12.7.2 Voltage Divider Bias

This is the most popular method. It uses a voltage divider (R_1, R_2) to set a precise positive voltage at the Gate.

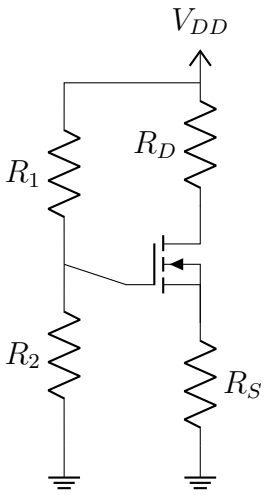


Figure 12.6: Voltage Divider Bias for E-MOSFET.

Analysis Steps: 1. Determine Gate Voltage:

$$V_G = V_{DD}\left(\frac{R_2}{R_1 + R_2}\right)$$

2. Write KVL for Gate-Source loop:

$$V_{GS} = V_G - I_D R_S$$

3. Substitute this into the characteristic equation:

$$I_D = k((V_G - I_D R_S) - V_T)^2$$

4. Solve the resulting Quadratic Equation for I_D.

Example 1: E-MOSFET Analysis

An E-MOSFET has $V_T = 3$V and $k = 0.4$ mA/V^2. It is biased using a voltage divider with $V_G = 5$V. The source is grounded ($R_S = 0$). Calculate I_D.

Solution: Since $R_S = 0$, $V_{GS} = V_G = 5$V.

$$I_D = k(V_{GS} - V_T)^2$$

$$I_D = 0.4(5 - 3)^2$$

$$I_D = 0.4(2)^2 = 0.4(4) = 1.6 \text{ mA}$$

12.8 CMOS Technology

Complementary MOS (CMOS) combines N-channel and P-channel E-MOSFETs on the same chip. It is the basis for almost all digital logic circuits (Inverters, Gates, Microprocessors).

12.8.1 The CMOS Inverter

- Configuration: A P-MOSFET connects the output to V_{DD} (Pull-up). An N-MOSFET connects the output to Ground (Pull-down). Their Gates are tied together as Input.
- Operation:
  - Input LOW (0V): NMOS is OFF ($V_{GS} = 0 < V_T$). PMOS is ON. Output is pulled HIGH (V_{DD}).
  - Input HIGH (V_{DD}): NMOS is ON ($V_{GS} > V_T$). PMOS is OFF. Output is pulled LOW (Gnd).
- Power: One transistor is always OFF. Therefore, in a static state, **Zero Current Flows** from supply to ground. Power is consumed only during the brief switching moment. This high efficiency is why your smartphone battery lasts all day despite having billions of transistors.

12.9 Handling Precautions

The insulating SiO_2 layer is extremely thin. Electrostatic Discharge (ESD) from your finger can build up thousands of volts, easily punching through the oxide and destroying the transistor permanently.

Safety Rules

- Always transport MOSFETs in anti-static foam.
- Wear a grounded wrist strap when handling components.
- Never touch the gate terminal before the source/drain are grounded.

12.10 Chapter Summary

1. MOSFETs use an Insulated Gate, resulting in ultra-high input impedance.
2. D-MOSFETs have a physical channel and operate in both depletion and enhancement modes.
3. E-MOSFETs have no channel; they are Normally OFF and require $V_{GS} > V_T$.
4. Current follows a square-law relationship: $I_D = k(V_{GS} - V_T)^2$.
5. CMOS logic uses complementary pairs for high efficiency.

12.11 Review Questions

12.11.1 Part A: Conceptual Questions

1. Explain the function of the Silicon Dioxide layer. Why does it result in infinite input impedance?
2. Differentiate between Depletion Mode and Enhancement Mode operation. Can a JFET operate in Enhancement mode? Why or why not?
3. Describe the formation of the ”Inversion Layer” in an N-Channel E-MOSFET.
4. Why is CMOS technology preferred for battery-operated devices?
5. What is ESD, and why are MOSFETs particularly susceptible to it?
6. Sketch the V-I transfer curve for an Enhancement type MOSFET. What happens when $V_{GS} < V_T$?
7. Why do E-MOSFETs require a positive gate voltage to turn ON (N-channel), whereas JFETs require a negative gate voltage to turn OFF?
8. Explain the difference between the physical structure of a D-MOSFET and an E-MOSFET.
9. What is the main advantage of Voltage Divider bias over Feedback bias for MOSFETs?
10. Draw the circuit diagram of a CMOS inverter.

12.11.2 Part B: Advanced Calculations & Analysis

11. **D-MOSFET Calculation:** A device has $I_{DSS} = 10$ mA and $V_P = -4$ V.
    (a) Calculate I_D at $V_{GS} = -2$V.
    (b) Calculate I_D at $V_{GS} = +2$V. (Note the positive input).
12. **E-MOSFET Constant (k):** A datasheet for an E-MOSFET lists $I_{D(on)} = 5$ mA at $V_{GS(on)} = 6$ V and $V_T = 3$ V.

(a) Calculate the constant k.

(b) Determine the current I_D if $V_{GS} = 8$V.

13. **Feedback Bias Analysis:** An E-MOSFET ($k = 0.2$ mA/V^2, $V_T = 2.5$ V) is biased using Feedback Biasing (R_G connects Drain to Gate). $V_{DD} = 15$V and $R_D = 2k\Omega$.

    (a) Show that $V_{GS} = V_{DS}$.

    (b) Write the KVL equation for the drain loop.

    (c) Calculate the resulting drain current I_D. (Requires solving quadratic).

14. **Voltage Divider Bias:** An E-MOSFET circuit has $V_{DD} = 24$V, $R_1 = 10M\Omega$, $R_2 = 5M\Omega$, $R_S = 1k\Omega$.

    (a) Calculate the Gate Voltage V_G.

    (b) If the transistor turns on and conducts 2 mA, calculate the actual V_{GS}.

    (c) Is the transistor ON if $V_T = 4$V?

15. **Transconductance Derivation:** Derive the formula for the transconductance g_m of an E-MOSFET by differentiating the current equation $I_D = k(V_{GS} - V_T)^2$. Then, calculate g_m for the device in Question 12 when $V_{GS} = 6$V.